

Grasshopper: Leaping into machine learning and quantitative analysis with Google Cloud

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About Grasshopper



By complementing
physical infrastructure
Google Cloud, Grasshopper
gained the flexibility to
support quantitative
development and
launch an innovative
media chat service in
finance; and leverage
powerful machine learning
and artificial intelligence

About Grasshopper

Headquartered in Singapore, Grasshopper is a leading technology-based trading company that typically completes thousands of trades a day and transacts more than \$500 billion in notional value of trades each month for clients.

Industries: Financial Services &
Insurance

Location: Singapore

Google Cloud results

- Provides quantitative developers with flexible, scalable CPU resources, enabling them to focus on research rather than infrastructure administration
- Allows leadership to reduce the financial, administrative, and manpower cost of infrastructure investment while achieving similar output as competitors
- Shortens development time and reduces

Enabling leading tech base firm - over \$500B in notional value monthly and efficiently develop new business lines.

Cloud Bigtable (<https://cloud.google.com/bigtable/>)
(<https://cloud.google.com/ai-platform/>),

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine
(<https://cloud.google.com/compute/>)

AI Platform (<https://cloud.google.com/ai-platform/>)

TensorFlow (<https://www.tensorflow.org/>)

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- Enables

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John Lin started his career as an open outcry trader. When the Singapore Exchange trading pits closed in 2005, John founded Grasshopper (<https://grasshopperasia.com/>), an electronic trading firm.

The following year, John hired his first trader and started developing a custom algorithmic trading program. By 2008, Grasshopper had become a successful high-frequency, algorithmic trading firm, and established itself in a highly competitive market.

The Singapore-based business now typically completes thousands of trades a day and transacts more than \$500 billion in notional value of trades each month for clients. Grasshopper trades a range of products, including listed derivatives, equities, foreign exchange, commodities, options, and exchange-traded funds, and is a designated market

maker – a partner that is always available to buy or sell products.

Tan T-Kiang, who joined Grasshopper as Chief Technology Officer in 2011, says despite growing to a workforce of 65 people, the business maintains a lean culture of innovation, performance, and delivery. "When I started with John, we had about 20 people," says Tan. "Even as we've grown and put more structures in place, we've always made maintaining our culture a top priority, and will continue to do so."

Grasshopper prides itself on its low-ego ethos, which guides it to base decisions on what the market tells it, rather than on pride or fear. By adding cultural values of curiosity, empowerment, and flexibility to the mix, the firm is able to boldly explore new terrain without worrying about making mistakes.

This approach has enabled an open outcry trader to become head of a cutting-edge technology firm, and led Grasshopper to embrace cloud computing with Google, while making forays into chat applications and cryptocurrencies.

From a single server

Grasshopper started operations in 2005 with a single high-end desktop as a server – a now-inconceivable notion. As trading volumes grew, the business added more servers and set up a dedicated server room. Quickly, the firm ran out of space and turned to a data center for rack space to

hold its servers and associated equipment. Further growth led to data centers in different countries and dedicated leased lines between locations. "We pushed ourselves to the point where we're competing with the best electronic trading firms in the world," says Tan.

The emergence of the cloud in Singapore and other markets initially made only a few ripples at Grasshopper. The business had strict latency, performance, and connectivity requirements, and was highly protective of its core trading systems. However, in 2015, Grasshopper began trialing the cloud to help secure non-critical operations.

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Grasshopper*

New cloud opportunities

Tan and his team identified opportunities to use cloud infrastructure to power development of the quantitative algorithms and signals that support Grasshopper's trading teams. "Our quantitative developers need scalable CPU resources to process jobs quickly and complete research in a timely way," says Tan. "We also wanted to improve our machine learning and artificial intelligence capabilities."

"The cloud and machine learning would allow us to avoid the growing pains of competitors that recruited dozens of quantitative developers and number-crunchers," he adds. "We could expand by combining the skills of our quantitative developers with automation and insights through machine learning in particular."

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Google Cloud for compute and ML

Grasshopper selected Google Cloud
(<https://cloud.google.com/>) to provide compute,
analytics, storage, and machine learning services to
complement its existing physical infrastructure.
Cloud Storage (<https://cloud.google.com/storage/>) holds
relatively inactive data, while BigQuery

(<https://cloud.google.com/bigquery/>) warehouses processes active data and runs some testing query jobs. Cloud Machine Learning Engine (<https://cloud.google.com/ml-engine/>) powers large machine learning tasks, while Compute Engine (<https://cloud.google.com/compute/>) virtual machines run services not sensitive to latency issues.

"With Google Cloud, we can avoid increasing the size of our infrastructure and support team," says Tan. "But more importantly, we can remove quantitative developers' concerns about having to calculate the CPU resources they may need for a job and asking the infrastructure team to purchase or provision servers."

"With a physical infrastructure, they may be concerned about asking for too many resources or too few, or risking the senior leadership team thinking they are spending money unwisely," he adds. "The scalability and pay-for-what-you-use pricing of Google Cloud means quantitative developers can simply request budget and focus on what is important, which is doing the research. And from a leadership point of view, we wanted the cloud to free the minds of our staff so they could take us to where our competitors were – without Grasshopper having to make the same investments as its rivals."

TensorFlow and Bigtable key

To meet its machine learning requirements, the business turned to the Tensorflow (<https://www.tensorflow.org/>) open source machine

learning framework, originally developed within the Google AI organization, and the Cloud Bigtable (<https://cloud.google.com/bigtable/>) NoSQL database service, which is optimized for large analytical and operational workloads.

"The result for us is a distilled dataset that we push into our own servers, where another machine learning bot or process listens to live market data and makes decisions while we trade," explains Tan.

With Google Cloud and its physical infrastructure, Grasshopper has established itself as a key provider of liquidity in equity and futures markets around the globe. The business is a designated market maker on the SGX, JPX (the Japan Exchange), and the CME Group's options and futures exchange.

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Location-agnostic cryptocurrency exchange connectivity

Grasshopper has also launched Tilde, its digital asset division that provides over-the-counter cryptocurrency executions and market making services for tokens and exchanges. The business uses Cloud Storage for caching and cold data storage, Compute Engine virtual machine instances to provide server capacity, and BigQuery to handle data involved in transaction dates plus one day events.

"For our cryptocurrency services, most exchanges are mostly internet exchanges now, although some are starting to offer co-location," explains Tan.

"Google Cloud allows us to connect to exchanges in locations where the firm does not have existing infrastructure, without paying too much of a penalty in terms of latency."

With Google Cloud, Grasshopper is leaping into the future of electronic trading and is poised to further expand into new markets, with more value-added offerings.

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